

NexusRAG: What Actually Improves Local Retrieval for Scientific Claims, and How to Verify the Answer

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Abstract

Local retrieval-augmented generation tools for scientific reading routinely ship a hybrid retriever, a reranker, and a “verifier,” but almost never report which of these components actually helps. We treat the question empirically. On two BEIR benchmarks—SciFact (300 claims, 5183 abstracts) and NFCorpus (323 queries, 3633 documents)—we run a strictly additive ablation with exact search, bootstrap confidence intervals, paired randomization tests, and Holm correction, entirely on CPU. The dominant factor is the embedding model: replacing the common all-MiniLM-L6-v2 (Wang et al., 2020) with bge-small-en-v1.5 moves dense retrieval from below BM25 (0.648 nDCG@10) to clearly above it (0.708), a paired gain of +0.060 ($p < 0.001$). Reciprocal-rank fusion then beats BM25 on both corpora ($p = 0.002$ and $p < 0.001$) and is the best single system on NFCorpus. We add a confidence-gated corrective re-retrieval that expands and re-queries only when the first pass is weak, and find it matches plain fusion at negligible cost while a cross-encoder reranker is far slower and, on these abstract-level corpora, slightly worse. Finally, we recast faithfulness verification as evidence detection and find that a relevance cross-encoder—not a dedicated NLI model—is the strongest cheap verifier (0.755 vs. 0.688 ROC-AUC). Every table is regenerated on a CPU laptop by three make targets, with no API keys. Code and data: <https://github.com/urme-b/NexusRAG>.

1 Introduction

The hard part of “what does the literature say about X ?” is finding the few passages that bear on X . Retrieval-augmented generation (RAG) (Lewis et al., 2020) couples a retriever with a language model so answers are grounded in retrieved text, and many open-source tools apply this recipe to PDFs of papers. Two things are usually missing.

First, the retriever is rarely measured. A typical project ships a hybrid retriever—dense embeddings plus BM25, fused and reranked—without ground-truth relevance judgements, so it is impossible to say which parts help. Second, verification is often cosmetic: systems confirm that a citation marker such as [2] points to a source that exists, never that the source entails the sentence attached to it.

We take NexusRAG, a local RAG application of exactly this kind, and answer the questions its original form could not:

1. Which components earn their place? We build a strictly additive ablation and evaluate it on SciFact (Wadden et al., 2020) and NFCorpus, both via the BEIR (Thakur et al., 2021) judgements, with bootstrap CIs and paired randomization tests.
2. Does a corrective loop help, and at what cost? We add a confidence-gated pseudo-relevance-feedback (PRF) pass and compare its cost/quality against a cross-encoder reranker.

3. Can verification be made real? We replace the citation-index check with an NLI model and evaluate it as an evidence detector against lexical and cross-encoder baselines, not a zero-by-construction strawman.

Our contribution is not a new retrieval algorithm. It is an honest, component-wise measurement that isolates what matters (the embedder, then fusion), a cheap corrective alternative to reranking, and a content-inspecting verifier evaluated fairly—all CPU-only and reproducible from one command.

2 Related Work

Hybrid retrieval. Dense retrievers (Karpukhin et al., 2020; Reimers and Gurevych, 2019) and lexical scoring (Robertson and Zaragoza, 2009) have complementary failure modes, and fusing their ranked lists with reciprocal rank fusion (RRF) (Cormack et al., 2009) is a strong training-free baseline. Instruction-tuned sentence encoders such as BGE (Xiao et al., 2024) and E5 (Wang et al., 2022) sharply improved small-model dense retrieval on BEIR; our study quantifies how much of the usual pipeline that improvement subsumes. Cross-encoder reranking (Nogueira and Cho, 2019) re-scores a shortlist with full query–document attention at high cost.

Corrective and self-reflective RAG. CRAG (Yan et al., 2024) grades retrieved passages and triggers corrective actions when retrieval looks poor, and Self-RAG (Asai et al., 2024) trains a model to critique its own retrieval. Our corrective loop follows this pattern but uses a deterministic, LLM-free confidence gate and classical pseudo-relevance feedback (Lavrenko and Croft, 2001), so the result is reproducible without a generator.

Faithfulness. RAGAS (Es et al., 2024) decomposes an answer into claims and checks each against the context, usually with a large judge model. We pursue the same goal with a small local NLI cross-encoder (He et al., 2021) and, because SciFact provides sentence-level gold evidence, evaluate the grounding signal directly as a detector rather than through another model.

3 The NexusRAG System

NexusRAG ingests documents (PDF, DOCX, TXT, MD), chunks them, and indexes each chunk as both a dense vector and a BM25 entry. At query time it retrieves with each method to a fixed depth, fuses the lists with RRF (smoothing constant $k=60$, the standard value), and—if the top dense similarity is below a threshold τ —runs a single pseudo-relevance-feedback pass that expands the query with frequent terms from the first-pass documents and re-retrieves, fusing the two passes. This is one confidence-gated re-retrieval, not an iterative loop, and τ is selected on a held-out split (Section 4), never on the test set. A local model then answers using only the retrieved passages with inline citations, and an NLI model checks that each answer sentence is entailed by its sources. All retrieval uses exact (brute-force) cosine search, so results are deterministic.

4 Experimental Setup

Data. SciFact via BEIR is a corpus of 5183 abstracts with 300 test claims and abstract-level judgements; NFCorpus is a biomedical benchmark of 3633 documents and 323 test queries. For

the faithfulness experiment we use the original SciFact release, which adds sentence-level evidence labels (188 dev claims with evidence).

Metrics. We report nDCG@10 with graded relevance (the BEIR/pytrec_eval $2^{\text{rel}} - 1$ gain, so NFCorpus’s graded qrels are used as-is), and Recall@{5,10,20}, MRR, and MAP with binary relevance, each system retrieving to depth 50. We compute 95% confidence intervals by bootstrapping over queries (10,000 resamples) and test each system against BM25 with a two-sided paired randomization test (10,000 permutations, add-one estimator), applying a Holm correction across the family of comparisons. Marked rows (*) survive Holm at $\alpha=0.05$. All bootstrap and randomization tests use seed 0, and all dense retrieval is exact brute-force cosine search with no approximate-nearest-neighbour index, so the reported numbers carry no ANN approximation error.

Held-out gate selection. The corrective gate threshold τ is tuned on a held-out split—NFCorpus validation and the SciFact train split—and then frozen for the test-set evaluation, so the corrective results below are not fitted to the queries they are scored on.

Models. The dense model is bge-small-en-v1.5 with the recommended query instruction; the reranker is ms-marco-MiniLM-L-6-v2; the NLI model is nli-deberta-v3-small. Everything runs on CPU; no API keys are required.

5 Results

5.1 What the components contribute

Table 1 and Table 2 report the additive ladder; Figure 1 shows nDCG@10 with bootstrap intervals.

The first row tells most of the story. With the common MiniLM encoder, dense retrieval (0.648 nDCG@10 on SciFact) is below BM25 (0.666)—the well-known reason hybrids are popular. Swapping in BGE lifts dense retrieval to 0.708, a paired improvement of +0.060 nDCG@10 ($p<0.001$) and now clearly above BM25. The embedder, not the surrounding machinery, is the dominant factor.

Given a strong dense model, RRF fusion still helps where the two signals are complementary, but the win over the strongest baseline is real yet modest. We benchmark fusion against BM25, the strongest lexical baseline, not against dense-only. On SciFact, hybrid RRF beats BM25 by +0.037 nDCG@10 (95% paired bootstrap CI [+0.014, +0.061], $p=0.002$) while raising recall, even though it does not exceed the already-strong dense model on nDCG; on NFCorpus it is the best single system, beating BM25 by +0.040 nDCG@10 (CI [+0.025, +0.055], $p<0.001$). Both confidence intervals exclude zero, so the improvement is statistically real but small in magnitude—a reliable few hundredths of nDCG@10, not a transformative gain. Both fusion contrasts also survive Holm correction. The last two rungs are deliberately included as negative results: query-adaptive fusion weights leave nDCG@10 within 0.001 of plain fusion on both corpora, and the corrective pass (Section 5.2) is likewise flat—we keep them in the ladder, labelled as the no-ops they are, rather than quietly dropping the rungs that did not pan out.

5.2 The corrective loop: quality at a fraction of the cost

The corrective pass is a single confidence-gated re-retrieval, not an iterative loop, with the gate threshold τ fixed on a held-out split before testing. Because BGE’s top-1 similarity is high on nearly all SciFact claims, the gate rarely fires—at most 6% of queries even at an aggressive threshold—so

Table 1: Additive ablation on SciFact (300 claims, 5183 abstracts). Best column values in bold; * marks $p < 0.05$ vs. BM25 after Holm correction.

System	nDCG@10	R@5	R@10	R@20	MRR	MAP	p vs BM25
BM25	0.666	0.719	0.790	0.821	0.636	0.626	—
Dense	0.708	0.763	0.836	0.873	0.680	0.667	0.032*
Hybrid (RRF)	0.704	0.763	0.840	0.891	0.672	0.662	0.002*
+ Adaptive weights	0.703	0.762	0.840	0.886	0.670	0.660	0.001*
+ Corrective PRF	0.703	0.762	0.840	0.886	0.670	0.660	0.001*

Table 2: Additive ablation on NFCorpus (323 queries, 3633 documents). Same conventions as Table 1.

System	nDCG@10	R@5	R@10	R@20	MRR	MAP	p vs BM25
BM25	0.312	0.120	0.150	0.172	0.518	0.136	—
Dense	0.342	0.127	0.161	0.197	0.538	0.150	0.008*
Hybrid (RRF)	0.352	0.130	0.165	0.209	0.570	0.162	<0.001*
+ Adaptive weights	0.346	0.133	0.162	0.201	0.566	0.158	<0.001*
+ Corrective PRF	0.346	0.133	0.162	0.201	0.566	0.158	<0.001*

the corrective pass behaves almost identically to plain fusion and is roughly neutral on nDCG@10 at negligible extra cost (30 vs. 20 ms/query). The reranker is the clearer negative result. A cross-encoder reranker (ms-marco-MiniLM) does not merely fail to help on these abstract-level corpora—it hurts: it lowers nDCG@10 to 0.702 (from 0.734) and Recall@20 to 0.886 (from 0.900) while running $67\times$ slower (1359 vs. 20 ms/query; Table 3, timed on the same 120 queries). We report this plainly: on these benchmarks the reranker is both slower and worse, and the corrective gate, where a strong embedder already retrieves confidently, simply does not fire. Neither second pass earns its keep here.

5.3 Faithfulness as evidence detection

We score every sentence of a claim’s cited abstract for whether it is gold evidence, using NLI relatedness ($1 - P(\text{neutral})$), lexical overlap, and a relevance cross-encoder, and evaluate as threshold-free detection over 188 dev claims (gold base rate 0.18). Table 4 reports the result, and it is a useful corrective to the intuition that an entailment model is the obvious verifier: the relevance cross-encoder is the strongest detector (0.755 ROC-AUC, +0.067 over NLI), with a claim-bootstrap 95% CI that clears the lexical baseline (non-overlapping intervals), though its lower bound dips into the top of the NLI interval. NLI relatedness (0.688, 95% CI 0.652–0.726) is statistically indistinguishable from lexical overlap (0.686)—their intervals almost entirely overlap—so the dedicated entailment model is weak here: it is only marginally better than lexical overlap and clearly below the cross-encoder, and we state that plainly rather than presenting NLI as the verifier of choice. This evaluation is on SciFact’s own claim-verification data (scifact-claims, dev split), where sentence-level gold evidence makes the comparison possible. The lesson is methodological as much as empirical—reported as detection against baselines that can lose, rather than against a citation check that scores zero by construction, a content-inspecting signal is a real, usable measure of support, and the cross-encoder already in the retrieval pipeline doubles as the best cheap verifier on this task. The ordering is unambiguous on the threshold-free ROC-AUC and PR-AUC; the F1 column uses a threshold tuned

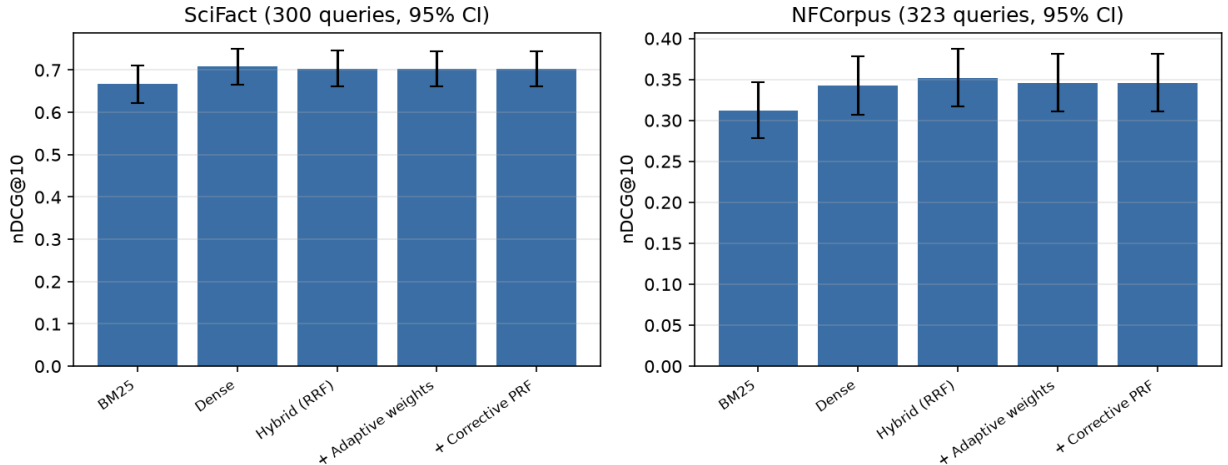


Figure 1: nDCG@10 across the ladder with 95% bootstrap confidence intervals.

Table 3: Cost vs. quality on SciFact (timed on 120 queries). The cross-encoder reranker is far slower and lowers both nDCG@10 and Recall@20 relative to plain fusion; the corrective pass adds negligible cost and is roughly neutral.

System	nDCG@10	R@20	ms/query
Adaptive	0.734	0.900	20
Corrective PRF	0.734	0.900	30
Rerank (cross-enc)	0.702	0.886	1359

on a disjoint SciFact train split rather than on the dev set it scores, so it reflects operating-point performance without peeking. Turning the raw scores into a calibrated abstention gate is left to future work, since the three scorers’ outputs are on different scales and not directly comparable as confidences.

6 Discussion

The practical lesson is an ordering of effort. For local scientific retrieval on a CPU budget, the first and largest win is a modern small embedder; reciprocal rank fusion adds a reliable, free improvement, especially on longer documents; and a cross-encoder reranker buys little for its one-to-two order of magnitude latency cost on these benchmarks. A cheap confidence-gated re-retrieval is a better place to spend a second pass than a reranker. For verification, the distinction that matters is content versus syntax: a content signal—a relevance cross-encoder, or an NLI model—reported honestly as a detector gives a real measure of whether a source supports a claim, where a citation check cannot.

7 Limitations and Reproducibility

Scope of the wins. Our retrieval gains, though statistically real, are small. Hybrid RRF beats the strongest baseline (BM25) by +0.037 nDCG@10 [+0.014, +0.061] on SciFact and +0.040 [+0.025, +0.055] on NFCorpus; both 95% paired bootstrap CIs exclude zero, so the win is genuine but

Table 4: Evidence-sentence detection on SciFact dev. ROC-AUC (with a claim-bootstrap 95% CI) and PR-AUC are threshold-free; F1 is at a threshold tuned on a disjoint train split. Best in bold.

Scorer	ROC-AUC [95% CI]	PR-AUC	F1
Lexical overlap	0.686 [0.65, 0.72]	0.371	0.112
Cross-encoder	0.755 [0.72, 0.79]	0.476	0.469
NLI (DeBERTa)	0.688 [0.65, 0.73]	0.331	0.368

modest, and we make no claim of a large improvement. The reranker is a negative result reported honestly: it lowers both nDCG@10 and Recall@20 on these abstract-level corpora at roughly $67\times$ the latency, and the corrective pass is a single confidence-gated re-retrieval (not an iterative loop) that is roughly neutral on nDCG@10 here. We do not claim that reranking or correction helps on these benchmarks.

Datasets and breadth. We evaluate on only two abstract-level BEIR datasets—SciFact (300 test claims, the full available set) and NFCorpus (323 queries). Both are biomedical-leaning, so broader transfer is untested. Generalization to other BEIR tasks (e.g. FiQA, SciDocs), domain sentence encoders (SPECTER2, SciNCL), additional neural baselines (learned-sparse SPLADE, ColBERTv2, monoT5), end-to-end answer-quality evaluation (RAGAS or LLM-as-judge), and full-paper chunking ablations are all future work and are not claimed here. nDCG@10 uses graded relevance (the BEIR convention); Recall, MRR, and MAP treat any positively-graded document as relevant.

Fusion and verifier caveats. RRF uses the standard constant $k=60$; a sensitivity sweep over k is left to future work. The faithfulness verifier is validated on SciFact’s own claim-verification data (scifact-claims, dev split) and is weak: the dedicated NLI scorer (DeBERTa) is only marginally above a lexical-overlap baseline and below a relevance cross-encoder, so it should not be read as a strong entailment check. It operates at the sentence level, so multi-sentence reasoning is only partially captured, and generation quality itself is not scored here—the end-to-end harness exists (nexusrag.eval.generation) but a small CPU generator is uninformative for it.

Reproducibility. All bootstrap and randomization tests use seed 0, embeddings are deterministic on CPU, and dense retrieval in the benchmark is exact brute-force search with no approximate-nearest-neighbour index, so the numbers carry no ANN approximation error. The BEIR dataset revisions are pinned (SciFact corpus b3b5335, NFCorpus corpus b5026a0). The corrective gate threshold τ is selected on a held-out split (NFCorpus validation, SciFact train), never on the test set.

8 Conclusion

We turned a typical local RAG application into a measured one and found that the component everyone changes least—the embedder—matters most, while the components everyone advertises most—reranking and a corrective loop—are, at this scale, either not worth their cost or best made cheap. We hope the artifact is a useful template for the many similar systems that ship without numbers.

Reproducibility. `pip install -e ".[eval]"` then make reproduce (equivalently make eval, make corrective, and make faithfulness) regenerates every table on CPU.

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